

Final QEA Project Presentation

BestBuy

By: Robert Lopez, Priya Sachin Bapte, Tata Abule Othow

2025

Contents / Agenda

1. Introductions
2. Technology used
3. Scenarios

Your name

- Robert Lopez
 - Graduated with a B.A in Computer Science
 - Location: New York
- Tata Abule Othow
 - Graduated with Bachelor of Computer Science
 - Location: Halifax, NS
 - Hobbie: Summer walk, Chess
- Priya Sachin Bapte
 - Graduated with Bachelor of Electronics engineering
 - Location: Edison, NJ

Skills learnt / Technology used

1. Agile and Test automation
2. Jira
3. Web UI and datasource
4. Automation Concepts, Selenium configuration, WebDriver Basics +Selenium with TestNG
5. Cypress
6. MongoDB
7. TOSCA
8. Digital technologies
9. A.I

What is Cucumber, Gherkin & Step Definitions?

Cucumber

- Supports Behavior-Driven Development for automation testing of web applications.
- Tests are Written in a simple Scenario form.

Gherkin

- Uses plain English Language for writing the test cases
- Uses Keywords :**Feature, Scenario, Given, Then, And**

Step Definitions

- A step definition is a method with an expression that links it to one or more Gherkin steps in the scenario

First feature

Feature: Searching on BestBuy

@T1 #robert

Scenario: searching for a product

Given I am on the BestBuy home page

When I close the add modal and search for "macbook pro"

Then one of the laptops listed should be 14 24GB Memory and 512GB SSD

@T1 #robert

Scenario: adding product to cart

Given I perform the above search

When I click the "Add to Cart" button next to the laptop

Then I should see a modal window with the cart subtotal

@T2 #Tata

Scenario: checking product is in the cart page

Given I am on the Best Buy modal page

When I click on go to cart

Then I navigate to the laptop and the order summary

@P2 #Tata

Scenario: Remove item from cart

Given I am on the Best Buy Cart page

When I remove an item from the cart

Then I verify that the item is removed from the cart

Scenario: Clean up

Then I close the browser



Gherkin Syntax



Steps (Given, When, Then)

Scenario 1:

Search for MacBook Pro

Scenario 1 – Requirements and Gherkin Syntax

- Given I am on the BestBuy home page
- When I close the add modal and search for “macbook pro”
- Then one of the laptops listed should be 14 24GB Memory and 512GB SSD

```
@T1 #robert
```

```
Scenario: searching for a product
```

```
    Given I am on the BestBuy home page
```

```
    When When I close the add modal and search for “macbook pro”
```



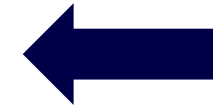
```
    Then one of the laptops listed should be 14 24GB Memory and 512GB SSD
```


Scenario 1 - BestBuySteps.java Step Definitions

```
WebDriver driver = WebDriverManager.getDriver(); 15 usages

@Given("I am on the BestBuy home page") @Lopez Garcia
public void i_am_on_the_best_buy_home_page() {

    // Write code here that turns the phrase above into concrete actions
    driver.get("https://www.bestbuy.com/");
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    try {
        WebElement usaFlag = wait.until(ExpectedConditions.visibilityOfElementLocated(By.className("us-link")));
        // If the page contains flags, click on the USA flag
        // If the page does not contain flags, this step will be skipped
        if(usaFlag.isDisplayed()){
            usaFlag.click();
            System.out.println("USA flag clicked successfully.");
        }
    } catch (TimeoutException e) {
        System.out.println("USA flag not found, continuing without clicking.");
    }
}
```



Step Definition – the implementation of the Gherkin step

Scenario 1 - BestBuySteps.java Step Definitions (cont.)

```
@When("When I close the add modal and search for \"macbook pro\"")  @Lopez Garcia *
public void when_i_close_the_add_modal_and_search_for_macbook_pro() {
    // Write code here that turns the phrase above into concrete actions

    System.out.println("When I close the add modal and search for \"macbook pro\"");
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    // wait for the input element to be visible
    wait.until(ExpectedConditions.visibilityOfElementLocated(By.className("store-display-name")));
    WebElement inputBar = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("autocomplete-search-bar")));
    // send keys to the input element
    inputBar.sendKeys(...keysToSend: "macbook pro");
    // find the search button and click it
    WebElement searchButton = driver.findElement(By.id("autocomplete-search-button"));
    // click the search button
    searchButton.click();
}
```

Scenario 1 - BestBuySteps.java Step Definitions (cont.)

```
@Then("one of the laptops listed should be 14 24GB Memory and 512GB SSD") @Lopez Garcia *
public void one_of_the_laptops_listed_should_be_14_24gb_memory_and_512gb_ssd() {
    // Write code here that turns the phrase above into concrete actions
    System.out.println("one of the laptops listed should be 14 8GB Memory and 256GB SSD");
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    // wait for the laptop with the specified specs to be visible
    WebElement macBook = wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath(xpathExpression: "//div[@class = 'sku-block'][contains(., '16') and contains(., '24GB Memory') and " +
        "contains(., '512GB SSD') and ./button[contains(., 'Add to cart')]]")));
    // assert that the laptop with the specified specs is found
    Assert.assertTrue(macBook.getText().contains("16"));
    Assert.assertTrue(macBook.getText().contains("24GB Memory"));
    Assert.assertTrue(macBook.getText().contains("512GB SSD"));
}
```

Scenario 2:

Adding to Cart

Scenario 2 – Requirements and Gherkin Syntax

- Given I perform the above search
- When I click the “Add to Cart” button next to the laptop
- Then I should see a modal window with the cart subtotal

```
@T1 #robert
```

```
Scenario: adding product to cart
```

```
Given I perform the above search
```

```
When I click the “Add to Cart” button next to the laptop
```

```
💡 Then I should see a modal window with the cart subtotal
```

Scenario 2 - BestBuySteps.java Step Definitions

```
@Given("I perform the above search") @Lopez Garcia *
public void i_perform_the_above_search() {
    // Write code here that turns the phrase above into concrete actions
    // Get the title of the search results page and verify it contains "macbook pro"
    String searchResult = driver.findElement(By.cssSelector("#promo-title")).getText();
    Assert.assertTrue(searchResult.contains("macbook pro"), message: "Search result does not contain 'macbook pro'");
}
```

Scenario 2 - BestBuySteps.java Step Definitions (cont.)

```
@When("I click the 'Add to Cart' button next to the laptop") @Lopez Garcia *
public void i_click_the_add_to_cart_button_next_to_the_laptop() {
    // Write code here that turns the phrase above into concrete actions
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    JavascriptExecutor js = (JavascriptExecutor) driver;
    WebElement macBook = wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath(xpathExpression: "//div[@class = 'sku-block'][contains(., '16') and contains(., '24GB Memory') and " +
        "contains(., '512GB SSD') and ./button[contains(., 'Add to cart')]]"))));
    System.out.println(macBook.getText());
    WebElement button = wait.until(ExpectedConditions.elementToBeClickable(macBook.findElement(By.xpath(xpathExpression: "./button[contains(., 'Add to cart')]]"))));
    js.executeScript(script: "arguments[0].scrollIntoView({block: 'center'});", button );

    try{
        button.click();
    }
    catch (StaleElementReferenceException e){
        System.out.println("Stale element reference exception caught");
        js.executeScript(script: "arguments[0].click();", button);
    }
}
```

Scenario 2 - BestBuySteps.java Step Definitions (cont.)

```
@Then("I should see a modal window with the cart subtotal") @Lopez Garcia *
public void i_should_see_a_modal_window_with_the_cart_subtotal() {
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));
    System.out.println("I should see a modal window with the cart subtotal");
    // wait for the cart subtotal to be visible
    WebElement cartSubtotal = wait.until(ExpectedConditions.visibilityOfElementLocated(By.className("cart-subtotal")));
    String actualText = cartSubtotal.getText();
    String expectedText = "Cart Subtotal";
    Assert.assertTrue(actualText.contains(expectedText), message: "Cart subtotal does not contain expected text: " + expectedText);
}
```


Scenario 3:

Add to Cart

Scenario 3 – Requirements and Gherkin Syntax

- Given I am on the BestBuy modal page
- When I click on go to cart
- Then I navigate to the laptop and the order summary

```
@T2 #Tata
```

```
Scenario: checking product is in the cart page
```

```
Given I am on the Best Buy modal page
```

```
When I click on go to cart
```

```
Then I navigate to the laptop and the order summary
```

Scenario 3 - BestBuySteps.java Step Definitions

```
// Scenario: checking product is in the cart page
@Given("I am on the Best Buy modal page") & Abule Othow +1 *
public void i_am_on_the_best_buy_modal_page() {
    // The code below checks if we are on the modal page
    System.out.println("I am in the Best Buy Modal page");
    String modalText = driver.findElement(By.className("added-to-cart")).getText();
    Assert.assertEquals(modalText, expected: "Added to cart", message: "Modal text does not match expected text");
}

@When("I click on go to cart") & Abule Othow *
public void i_click_on_go_to_cart() {
    // Try to click on the "Go to Cart" button in the modal
    try{
        Thread.sleep(millis: 5000);
        WebElement goToCart = driver.findElement(By.xpath(xpathExpression: "//*[@id='recs-interruptor-drawer-overlay-backdrop']/div/div[3]/div/div/div/div[2]"));
        goToCart.click();
    }catch (Exception e){
        e.printStackTrace();
    }
}

@Then("I navigate to the laptop and the order summary") & Abule Othow *
public void i_navigate_to_the_laptop_and_the_order_summary() throws InterruptedException {
    // Verify we are on the cart page and print the total amount
    String expUrl = "https://www.bestbuy.com/cart";
    String actUrl = driver.getCurrentUrl();
    Thread.sleep(millis: 5000);
    WebElement amountSummary = driver.findElement(By.xpath(xpathExpression: "//tr[.//span[text()='Total']]/td[starts-with(normalize-space(.), '$')]"));
    System.out.println("Total: "+ amountSummary.getText());
    Assert.assertEquals(actUrl, expUrl);
}
```

Scenario 4:

Remove Item from cart

Scenario 4 – Requirements and Gherkin Syntax

- Given I am on the BestBuy cart page
- When I remove an item from the cart
- Then I verify that the item is removed from the cart

```
@P2 #Tata
```

```
Scenario: Remove item from cart
```

```
    Given I am on the Best Buy Cart page
```

```
    When I remove an item from the cart
```

```
    Then I verify that the item is removed from the cart
```

Scenario 4 - BestBuySteps.java Step Definitions

```
// Scenario: Remove item from cart
@Given("I am on the Best Buy Cart page")  ⚙ Abule Othow
public void i_am_on_the_best_buy_cart_page() {
    // Below code checks if we are on the cart page
    System.out.println("I am on the Best Buy Cart page");
    String cartUrl = "https://www.bestbuy.com/cart";
    String actualUrl = driver.getCurrentUrl();
    Assert.assertEquals(actualUrl, cartUrl, message: "Not on the cart page");
}
```

```
@When("I remove an item from the cart")  ⚙ Abule Othow *
public void i_remove_an_item_from_the_cart() {
    // Write code here that turns the phrase above into concrete actions
    System.out.println("REMOVE ITEM FROM CART ");
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    // Wait for the cart to be clickable and click it
    try {
        // Wait for the remove button to be clickable and click it
        WebElement removeButton = wait.until(
            ExpectedConditions.elementToBeClickable(By.cssSelector("button.cart-item__remove"))
        );
        removeButton.click();
    } catch (Exception e) {
        throw new RuntimeException("Cart is not clickable: " + e.getMessage());
    }
}
```

Scenario 4 - BestBuySteps.java Step Definitions (cont.)

```
@Then("I verify that the item is removed from the cart") & Abule Othow *
public void i_verify_that_the_item_is_removed_from_the_cart() {
    // Write code here that turns the phrase above into concrete actions
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    // Wait for the cart to update and show "Your cart is empty" message
    Boolean emptyCartMessage = wait.until(ExpectedConditions.textToBe(
        By.cssSelector("h1.heading-5.page-heading__title"),
        value: "Your cart is empty"
    ));
    Assert.assertTrue(emptyCartMessage, message: "The message does not match. The cart is not empty.");
}
```

Scenario 5: Password Validation for BestBuy using Data Driven Testing

Password Testing Feature File

Scenario Outline: Password validation

@L3

Scenario Outline: Password Validations

Given start with the BestBuy home page
Then I click on Account button
And I see the panel with Create account button
When I click on the Create Account button
Then I should be navigated to Create Account page
When I enter <password> it should check for password validations <Valid>

Examples:

password	Valid	
<u>"QEA@team2"</u>	"true"	
"JohnDoe\$23"	"true"	
"john"	"false"	

Password Validation - Step Definitions

```
//Step to launch the browser
@Given("start with the BestBuy home page") @Bapte*
public void startWithTheBestBuyHomePage() throws InterruptedException {
    driver.get("https://www.bestbuy.com/");
    Thread.sleep( millis: 5000);
}

// Step to click on Account button
@Then("I click on Account button") @Bapte*
public void i_click_on_account_button() throws InterruptedException {
    // Wait for the Account button to be clickable
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(5));
    WebElement accountButton = wait.until(ExpectedConditions.elementToBeClickable(
        By.xpath( xpathExpression: "//span[@class='v-p-right-xxs line-clamp']"))));
    accountButton.click();
}

@Then("I see the panel with Create account button") @Bapte*
public void i_see_the_panel_with_create_account_button() throws InterruptedException {
    // Wait for the panel to be visible
    // create account button is displayed: Create Account

    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(5));
    WebElement element = wait.until(ExpectedConditions.visibilityOfElementLocated(
        By.xpath( xpathExpression: "//a[text()='Create Account']"))));
    Assert.assertTrue(driver.findElement(
        By.xpath( xpathExpression: "//a[text()='Create Account']")).isDisplayed());
}
```

Password Validation - Step Definitions

```
@When("I click on the Create Account button") @Bapte*
public void i_click_on_the_create_account_button() throws InterruptedException {
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(5));
    WebElement element = wait.until(ExpectedConditions.visibilityOfElementLocated(
        By.xpath(xpathExpression: "//a[text()='Create Account']")));
    element.click();
}

@Then("I should be navigated to Create Account page") @Bapte*
public void i_should_be_navigated_to_create_account_page() throws InterruptedException {
    String expected = "Best Buy: Create an Account";

    // Get the actual title of the page
    String actualUrl = driver.getTitle();
    Assert.assertEquals(actualUrl, expected);
}
```

Password Validation - Step Definitions

```
// Testing Positive and Negative password validations
@When("I enter {string} it should check for password validations {string}") Bapte*
public void i_enter_it_should_check_for_password_validations(String password, String valid) {
    WebElement passwordText = driver.findElement(By.id("fld-p1"));
    passwordText.sendKeys(password);
    passwordText.sendKeys(Keys.TAB);
    List<WebElement> passwordElement = driver.findElements(By.xpath
        ( xpathExpression: "//p[text()='Please enter a strong password.']"));

    if(valid.equals("true")){
        System.out.println("Valid = true " + passwordElement.isEmpty());
        Assert.assertTrue(passwordElement.isEmpty());
    }
    else {
        System.out.println("Valid = false " + passwordElement.isEmpty());
        Assert.assertTrue(passwordElement.getFirst().isDisplayed());
    }
}
```

Create an Account

Shopping for your business? [Create a business account.](#)

....



⚠ Please enter a strong password.

Scenario 6: Login Functionality

Create an account for Best Buy

Best Buy Create an account Gherkin syntax and step definition

Feature: User operations on BestBuy

@L1

Scenario: Creating a new user

Given start with the BestBuy home page

Then I click on Account button

And I see the panel with Create account button

When I click on the Create Account button

Then I should be navigated to Create Account page

And I can enter all of my account information

Then I should be navigated to my account page

```
//Step to launch the browser
@Given("start with the BestBuy home page") @Bapte
public void startWithTheBestBuyHomePage() throws InterruptedException {
    driver.get("https://www.bestbuy.com/");
    //Thread.sleep(5000);

    driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(2));
    List<WebElement> banners = driver.findElements(cookieBanner);
    if (!banners.isEmpty() && banners.getFirst().isDisplayed()) {
        // Try click accept; if intercepted, scroll and retry
        List<WebElement> buttons = driver.findElements(acceptButton);
        if (!buttons.isEmpty()) {
            try {
                buttons.getFirst().click();
            } catch (ElementClickInterceptedException e) {
                ((JavascriptExecutor)driver).executeScript("script: arguments[0].scrollIntoView(true);");
                buttons.getFirst().click();
            }
        }
        // wait for banner to vanish
        new WebDriverWait(driver, Duration.ofSeconds(5))
            .until(ExpectedConditions.invisibilityOfElementLocated(cookieBanner));
    }
}
```

BestBuy .java Step Definitions Cond

```
}  
// Step to click on Account button  
@Then("I click on Account button") @Bapte*  
public void i_click_on_account_button() throws InterruptedException {  
    // Wait for the Account button to be clickable  
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(5));  
    WebElement accountButton = wait.until(ExpectedConditions.elementToBeClickable(  
        By.xpath(xpathExpression: "//span[@class='v-p-right-xs line-clamp']"))));  
    accountButton.click();  
}
```

```
// Method to generate a unique email address  
String uniqueEmail() no usages @Bapte*  
{  
    String prefix = "testuser";  
    String domain = "@gmail.com";  
    long timestamp = System.currentTimeMillis();  
    return prefix + timestamp + domain;  
}
```

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Create an Account

 An account with this email already exists.
Please [sign in](#) or use a different email address.

Shopping for your business? [Create a business account.](#)

First Name

Kevin

Last Name

Tal

Email Address

kevintamalas@gmail.com

Password

.....



Confirm Password

.....



 Your passwords match!

Scenario 7: Login Functionality

Create an account for Staples.com

Scenario 7 – Gherkin Syntax – Create an account with Valid credentials

@L9

Scenario: Successfully register a new Staples user

Given I navigate to the Staples account creation page

When I enter a unique email

And I enter "Alexander" into the First Name field of staples

And I enter "Steve" into the Last Name field of staples

And I enter "7327121878" into the Phone Number field

And I enter "SecurePass123!" into the Password field of staples

And I select "Myself" for the shopping preference

And I click the Create Account button of staples

Then I should see a confirmation message or be redirected to My Account

Scenario 6 – StaplesRegisterSteps.java Step Definitions

```
@Given("I navigate to the Staples account creation page")
public void goToCreateAccountPage() throws InterruptedException {

    driver.get("https://www.staples.com/idm/com/createaccount");
    Thread.sleep(millis: 20000);
    handleIFrames();
}

private void handleIFrames() throws InterruptedException { 1 usage
    try { driver.switchTo().activeElement().sendKeys(Keys.ESCAPE); } catch (Exception ignored) {}
    ((JavascriptExecutor)driver).executeScript(
        script: "document.querySelectorAll('[role='dialog'],.modal,.popup,[data-modal='true']\").forEach(e=>{" +
            "e.style.setProperty('display','none','important');});"
    );
    Thread.sleep(millis: 1000);
    driver.switchTo().defaultContent();
}
```

Scenario 6 – StaplesRegisterSteps.java Step (contd.)

```
@When("I click the Create Account button of staples")
public void clickCreateAccount() throws InterruptedException {
    driver.findElement(By.id("submitIdmCreateForm")).click();
    Thread.sleep( millis: 2000);
}

@Then("I should see a confirmation message or be redirected to My Account")
public void verifyAccountCreated() {
    String expectedUrl="https://www.staples.com/idm/com/createaccount";
    String actualUrl = driver.getCurrentUrl();
    Assert.assertEquals(actualUrl, expectedUrl);
    driver.quit();
}
```



Your Staples account has been created.

You're ready to start shopping!

Shop now

You've also unlocked Easy Rewards benefits and will now earn points on your purchases!



1 point per \$1 spent

Negative Scenario 8: Invalid phone no.

Negative scenario – Staples register – Invalid phone

@L10

Scenario: Negative Test - Incorrect phone number format on Staples

Given I navigate to the Staples account creation page

When I enter a unique email

And I enter "Alexander" into the First Name field of staples

And I enter "Steve" into the Last Name field of staples

And I enter "1111111111" into the Phone Number field

Then I should see a incorrect phone number format error message

Negative scenario - Staples register

```
@When("I enter {string} into the First Name field of staples")
public void enterFirstName(String firstName) {
    driver.findElement(By.id("firstName")).sendKeys(firstName);
}

@When("I enter {string} into the Last Name field of staples")
public void enterLastName(String lastName) {
    driver.findElement(By.id("lastName")).sendKeys(lastName);
}

@When("I enter {string} into the Phone Number field")
public void enterPhoneNumber(String phone) {
    //driver.findElement(By.id("mobileNumber")).sendKeys(phone);
    typeSlowly(By.id("mobileNumber"), phone);
    WebElement phoneField = driver.findElement(By.id("mobileNumber"));
    phoneField.sendKeys(Keys.TAB);
}
```

Negative scenario - Staples register Step (contd.)

```
@Then("I should see a incorrect phone number format error message")
public void iShouldSeeAIncorrectPhoneNumberFormatErrorMessage() {
    // Write code here that turns the phrase above into concrete actions
    Assert.assertTrue(driver.findElement(
        by.xpath(xpathExpression: "//div[text()='Please enter a valid phone number.']").isDisplayed());
}
```

Create an account

Email*
testuser1756354397781@example.com

First name*
Alexander

Last name*
Steve

Phone number*
(111) 111-1111

! Please enter a valid phone number.

Scenario 9:

Cleanup

Scenario 8 – Requirements and Gherkin Syntax

- Then I close the browser

```
Scenario: Clean up  
    Then I close the browser
```

Scenario 8 – BestBuySteps.java Step Definitions

```
@Then("I close the browser")  👤 Lopez Garcia +1
public void i_close_the_browser() {
    // Write code here that turns the phrase above into concrete actions
    driver.quit();
}
```

Problems Encountered

- We tried to implement the login feature for BestBuy.com but found that they are blocking accounts from being logged in autonomously.
- Creating an account was also blocked unless done manually.

Create an Account

 **An account with this email already exists.**
Please [sign in](#) or use a different email address.

Shopping for your business? [Create a business account.](#)

First Name

Kevin

Last Name

Tal

Email Address

kevintamalas@gmail.com

Password

.....



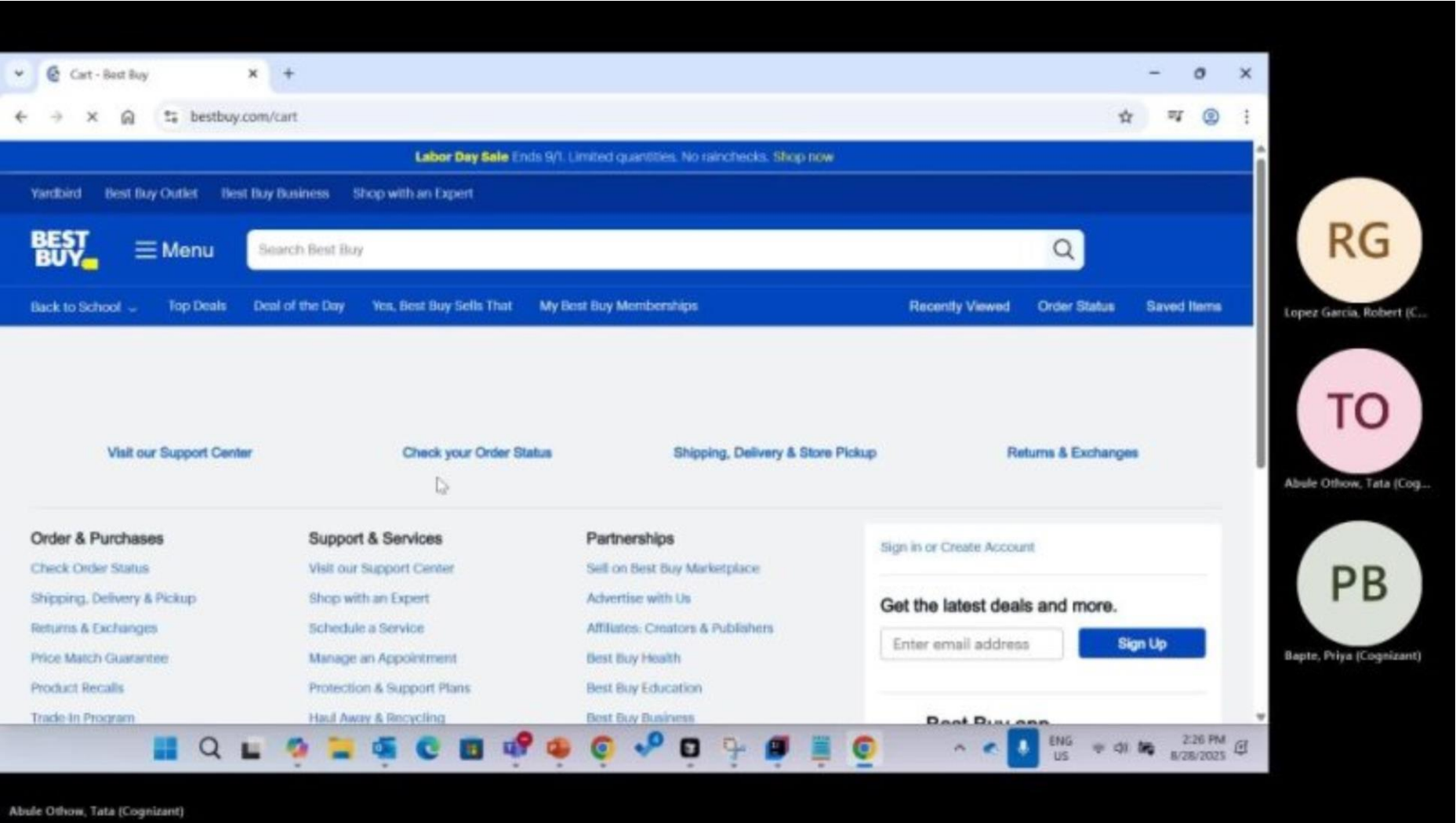
Confirm Password

.....



 Your passwords match!

Run-Through

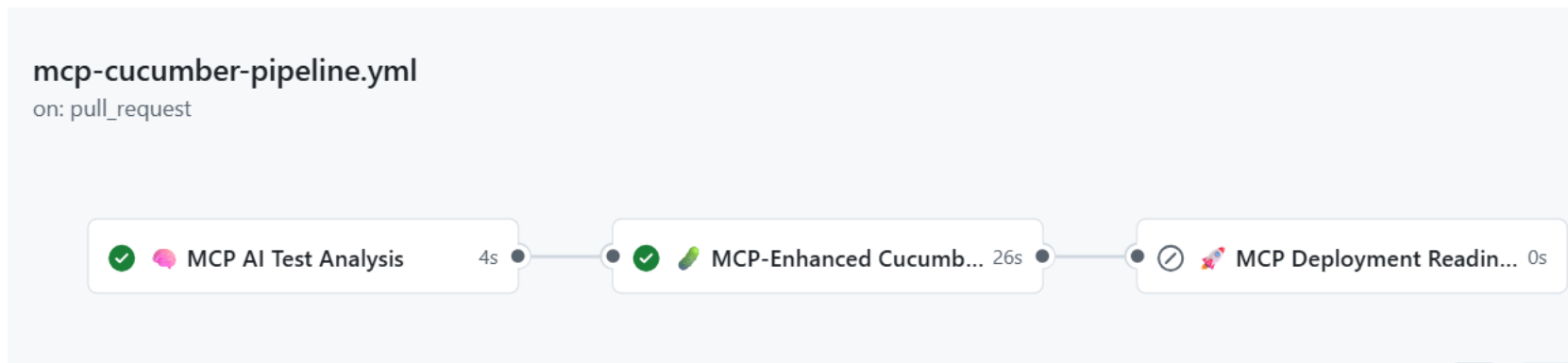


Test Reports

1 FAILED		9 PASSED	
90% passed 10 executed		4 hours ago last run	3 minutes 16 seconds duration
 Windows 11		 OpenJDK 64-Bit Server VM 22.0.2+9	 cucumber-jvm 7.14.0
Search with text or @tags You can search with plain text or Cucumber Tag Expressions to filter the output ☒ failed ☒ passed			
>  file:///C:/Users/2433393/OneDrive%20-%20Cognizant/Desktop/Cucumber-Capstone/src/test/resources/features/BestBuy.feature			
>  file:///C:/Users/2433393/OneDrive%20-%20Cognizant/Desktop/Cucumber-Capstone/src/test/resources/features/BestBuylogin.feature			

GitHub CI/CD Pipeline

GitHub Actions with MCP AI is a platform for creating a Continuous Integration/Continuous Delivery (CI/CD) environment. I built a CI/CD pipeline using Cursor AI, integrating GitHub Actions with the MCP AI concept to support intelligent testing, automated decision-making, and efficient deployment of Cucumber BDD test suites.



Checklist that included in the Capstone

Core Java Project

- Successful run through focusing on Cucumber Testing
- Incorporated an Object-Oriented Java Programming Concepts as required
- Appropriate data type choices and usage

☐ Maven Framework

- Use Maven as build management tool to successful manage library dependencies

☐ Cucumber Technology & Gherkin Syntax

- Explicit use of Cucumber technology and Gherkin language

☐ Git

- Appropriate use of git using push, add, and other commands

☐ Use of Given, When & Then

- Successful run through Given, When & Then in Gherkin Syntax for search, add cart, create a new user and password validations.

Interim Feedback

Capstone Summary

In our Capstone project, we implemented multiple automated test cases using **Cucumber BDD**, **Gherkin syntax**, and **Java-based step definitions**.

Key scenarios included :

- Searching for products (e.g., MacBook Pro)
- Adding and removing items from the cart
- Validating password strength using data-driven testing
- Creating user accounts on BestBuy and Staples
- Handling negative scenarios like invalid phone numbers
- Final cleanup and browser closure

Skills learnt / Technology used

1. Agile and Test automation
2. Jira
3. Web UI and datasource
4. Automation Concepts, Selenium configuration, WebDriver Basics +Selenium with TestNG
5. Cypress
6. MongoDB
7. Tosca
8. Digital technologies
9. A.I

Run-Through Video

Microsoft Teams

Call with Abule Othow, Tata (Cognizant)

2025-09-09 14:45 UTC

Recorded by

Bapte, Priya (Cognizant)

Interim Check in feedback

Based on the feedback given to us:

1. Implement a plugin extension for reporting where we are going to include use case traceability, and positive and negative test categorization.
2. We are also going to set up MCP with GitHub Copilot to help automate our CI/CD process.

1. Testing report

@Positive

8 passed

STATUS	TIMESTAMP	TESTNAME
Pass	12:32:05 PM	searching for a product Searching on BestBuy.searching for a product
Pass	12:32:36 PM	adding product to cart Searching on BestBuy.adding product to cart
Pass	12:32:43 PM	checking product is in the cart page Searching on BestBuy.checking product is in the cart page
Pass	12:32:59 PM	Remove item from cart Searching on BestBuy.Remove item from cart
Pass	12:33:00 PM	Password Validations User operations on BestBuy.Password Validations
Pass	12:33:18 PM	Password Validations User operations on BestBuy.Password Validations
Pass	12:33:45 PM	Successfully register a new Staples user User operations on BestBuy.Successfully register a new Staples user
Pass	12:34:56 PM	Clean up User operations on BestBuy.Clean up

Category

14

2 tests

@Negative

2 tests

@UC-102

1 tests

@phoneValidation

1 tests

@Negative

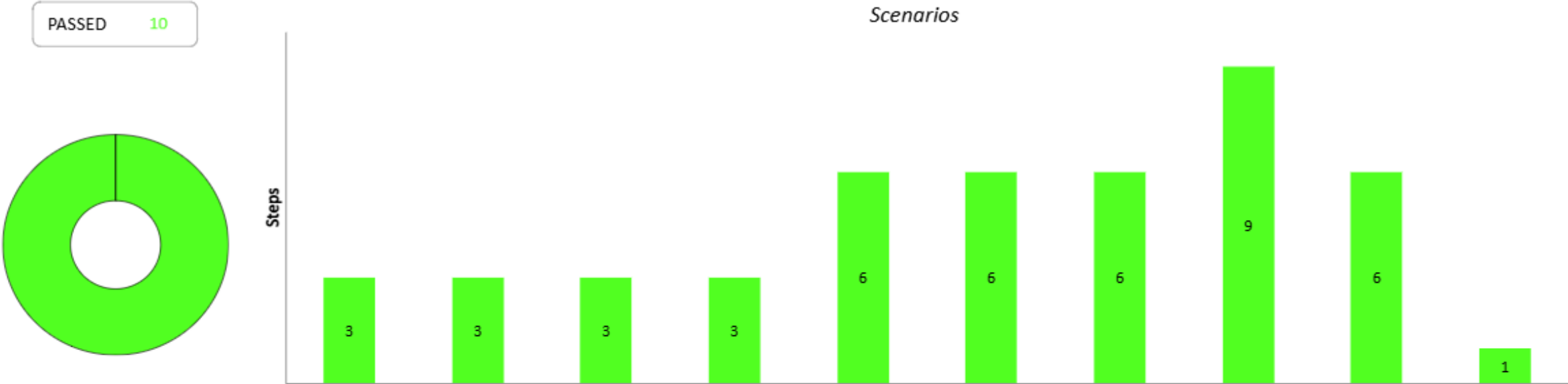
2 passed

STATUS	TIMESTAMP	TESTNAME
Pass	12:33:32 PM	Password Validations User operations on BestBuy.Password Validations
Pass	12:34:32 PM	Negative Test - Incorrect phone number format on Staples User operations on BestBuy.Negative Test - Incorrect phone number format on Staples

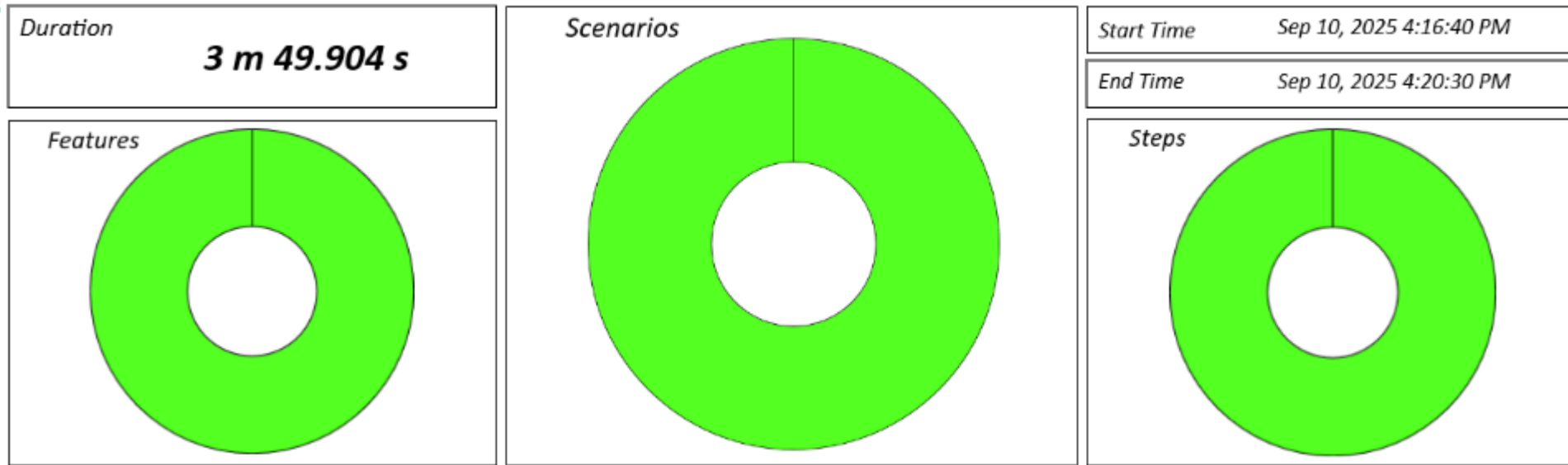
Test report use case

NAME	PASSED	FAILED	SKIPPED	OTHERS	PASSED %
@Login	1	0	0	0	100%
@UC-109	1	0	0	0	100%
@UC-106	1	0	0	0	100%
@UC-104	1	0	0	0	100%
@UC-108	1	0	0	0	100%
@UC-105	2	0	0	0	100%
@Negative	2	0	0	0	100%
@UC-102	1	0	0	0	100%
@phoneValidation	1	0	0	0	100%
@UC-103	1	0	0	0	100%
@UC-107	1	0	0	0	100%
@Positive	8	0	0	0	100%
@UC-101	1	0	0	0	100%
@PasswordValidation	3	0	0	0	100%

Detailed Report in Excel File



SCENARIO			FEATURE		STEPS			
NAME	STATUS	DURATION	NAME	STATUS	TOTAL	PASSED	FAILED	SKIPPED
searching for a product	PASSED	42.387 s	Searching on BestBuy	PASSED	3	3		
adding product to cart	PASSED	15.191 s	Searching on BestBuy	PASSED	3	3		
checking product is in the cart page	PASSED	21.898 s	Searching on BestBuy	PASSED	3	3		
Remove item from cart	PASSED	0.837 s	Searching on BestBuy	PASSED	3	3		
Password Validations	PASSED	21.358 s	User operations on BestBuy	PASSED	6	6		
Password Validations	PASSED	20.366 s	User operations on BestBuy	PASSED	6	6		
Password Validations	PASSED	17.839 s	User operations on BestBuy	PASSED	6	6		
Successfully register a new Staples user	PASSED	51.038 s	User operations on BestBuy	PASSED	9	9		
Negative Test - Incorrect phone number format on	PASSED	25.559 s	User operations on BestBuy	PASSED	6	6		
Clean up	PASSED	11.519 s	User operations on BestBuy	PASSED	1	1		



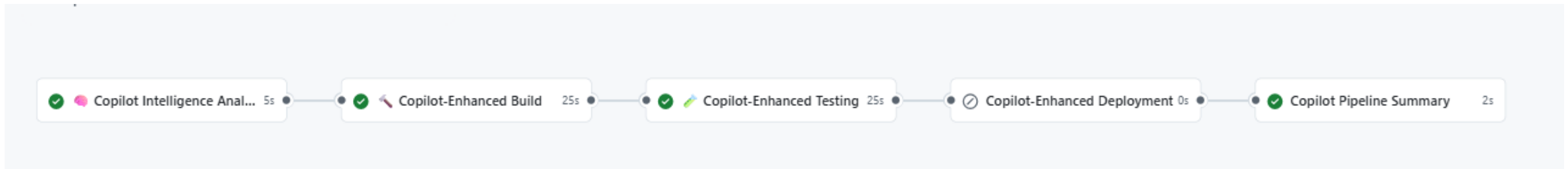
TAG NAME	FEATURE NAME	SCENARIO NAME	SCENARIO STATUS
@Positive	Searching on BestBuy	searching for a product	PASSED
		adding product to cart	PASSED
		checking product is in the cart page	PASSED
		Remove item from cart	PASSED
	User operations on BestBuy	Password Validations	PASSED
		Password Validations	PASSED
		Successfully register a new Staples user	PASSED
		Clean up	PASSED
@UC-101	Searching on BestBuy	searching for a product	PASSED
@UC-102	Searching on BestBuy	adding product to cart	PASSED
@UC-103	Searching on BestBuy	checking product is in the cart page	PASSED
@UC-104	Searching on BestBuy	Remove item from cart	PASSED
@UC-105	User operations on BestBuy	Password Validations	PASSED
		Password Validations	PASSED

2. MCP - GitHub CI/CD Pipeline

Cursor AI built an intelligent CI/CD pipeline using Model Context Protocol (MCP) that mimic GitHub Copilot's decision-making.

MCP is implemented through three core components:

- Context Analysis
- Protocol Selection: High – Comprehensive testing, Medium – Regression testing, Low – Smoke testing
- Automation Execution



Thank you

Any Questions?

Robert Lopez, Tata Abule, Priya Bapte